

SIRCSS-CTA

Lab 1: Topic models

1 Topic Models

This assignment uses the R package `uuml` with data and functionality to simplify coding. To install the packages just run the following:

```
install.packages("remotes")
remotes::install_github("MansMeg/IntroML", subdir = "rpackage")
install.packages(c("dplyr", "topicmodels", "tm", "zoo"))
```

What to submit

Submit a short report or notebook that includes:

- the code needed to reproduce your analysis;
- a convergence plot based on the log-likelihood values;
- a table with the top words for each topic;
- interpretations of two coherent topics;
- topic-over-time plots for the two interpreted topics;
- a short analysis where you change the number of topics, K .

We will now analyze the classical book *Pride and Prejudice* by Jane Austen using a probabilistic topic model. If you have not read the book, [here](#) you can read up on the story.

For this part of the assignment, [Griffiths and Steyvers \(2004\)](#) is the primary reference. I would also recommend reading [Blei \(2012\)](#) before starting with the assignment.

We will use a Gibbs sampler to estimate ten different topics occurring in *Pride and Prejudice* and study where they occur. A tokenized version of the book and a `data.frame` with stopwords can be loaded as follows:

```
library(uuml)
library(dplyr)
library(tidytext)
data("pride_and_prejudice")
data("stopwords")
```

1. As a first step, we will remove stopwords (common English words without much semantic information):

```
pap <- pride_and_prejudice
pap <- anti_join(
  pap,
  y = stopwords[stopwords$lexicon == "snowball",],
  by = "word"
)
```

2. Then we will remove rare words. Here we remove words that occur five times or fewer.

```
word_freq <- table(pap$word)
rare_words <- data.frame(
  word = names(word_freq[word_freq <= 5]),
  stringsAsFactors = FALSE
)
pap <- anti_join(pap, y = rare_words, by = "word")
```

3. Now we have a corpus we can use to implement a probabilistic topic model. We do this by using the `topicmodels` R package. As a first step we will compute a document term matrix using the `tm` package, where we treat each paragraph as a document. How many documents and terms (word types) do you have?

```
library(tm)
crp <- aggregate(
  pap$word,
  by = list(pap$paragraph),
  FUN = paste0,
  collapse = " "
)
names(crp) <- c("paragraph", "text")
s <- SimpleCorpus(VectorSource(setNames(crp$text, crp$paragraph)))
m <- DocumentTermMatrix(s)
dim(m)
```

4. To compute a topic model with ten topics, we use a Gibbs sampling algorithm. Below is an example of how we can run a Gibbs sampler for 2000 iterations. Run your topic model for 2000 iterations. Gibbs sampling is stochastic; the seed below makes your result reproducible.

```
library(topicmodels)
K <- 10
# Note: delta is beta in Griffiths and Steyvers (2004) notation.
control <- list(keep = 1, delta = 0.1, alpha = 1, iter = 2000)
set.seed(2026)
lda_fit <- LDA(m, k = K, method = "Gibbs", control)
```

5. In the `uuml` R package you have three convenience functions to extract Θ , Φ and the log-likelihood values at each iteration. This is the parameter notation used in Griffiths and Steyvers (2004).

```
library(uuml)
lls <- extract_log_liks(lda_fit)
theta <- extract_theta(lda_fit)
phi <- extract_phi(lda_fit)
dim(theta)
dim(phi)
length(lls)
```

6. As a first step, check that the model has converged by visualizing the log-likelihood over epochs/iterations. Does it seem like the model has converged?
7. Extract the 20 top words for each topic (i.e. the words with the highest probability in each topic). Choose two topics you find coherent/best (the top words seem to belong together). Interpret these two topics based on the storyline of the book. What have these two topics captured?

```
top_words <- terms(lda_fit, 20)
top_words
```

8. Visualize how these two topics evolve over the paragraphs in the book by plotting the θ parameters for each topic over time. Think of this as the timeline of the book. On the y-axis, you should plot θ_i for your chosen topic i and the x-axis should be the original paragraph number.

```
paragraph_ids <- as.integer(rownames(theta))
```

9. How do these two chosen topics evolve over the course of the book? If you want, you can take a rolling mean of the theta parameters to more easily show the changes in the topic over the book. *Hint!* Here `zoo::rollmean()` might be a good function to use.
10. Test to change the number of topics and do your own analysis of the novel when you feel you have a good number of topics.

References

- Thomas L. Griffiths and Mark Steyvers. Finding scientific topics. *Proceedings of the National Academy of Sciences*, 101(suppl 1):5228–5235, 2004.
- David M. Blei. Probabilistic topic models. *Communications of the ACM*, 55(4):77–84, 2012.